



CANDIDATE NAME

CT GROUP

17S7 ____

BIOLOGY

8876/02

Paper 2 Structured and Free-response Questions

27 August 2018

No additional materials are required.

2 hours

INSTRUCTIONS TO CANDIDATES

Write your **name** and **CT group** in the spaces provided at the top of this cover page.

SECTION A

This section contains **six** questions. Answer **ALL** questions in the spaces provided on the Question Paper.

SECTION B

This section contains **two** questions. Answer any **one** question in the spaces provided on the Question Paper.

Your answers must be in continuous prose, where appropriate.

A **NIL RETURN** is required for questions not answered.

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	/ 8
2	/ 8
3	/ 9
4	/ 8
5	/ 7
6	/ 5
7 / 8	/ 15
Final Mark	/ 60

This document consists of **20** pages.

Section A

Answer **all** the questions in this section.

QUESTION 1

Fig. 1.1 shows a goblet cell from the colon of a rat, which can be found scattered among the epithelial cells that line the intestinal tract. The goblet cell produces mucus that is subsequently packaged into mucigen granules before being secreted out of the cell.

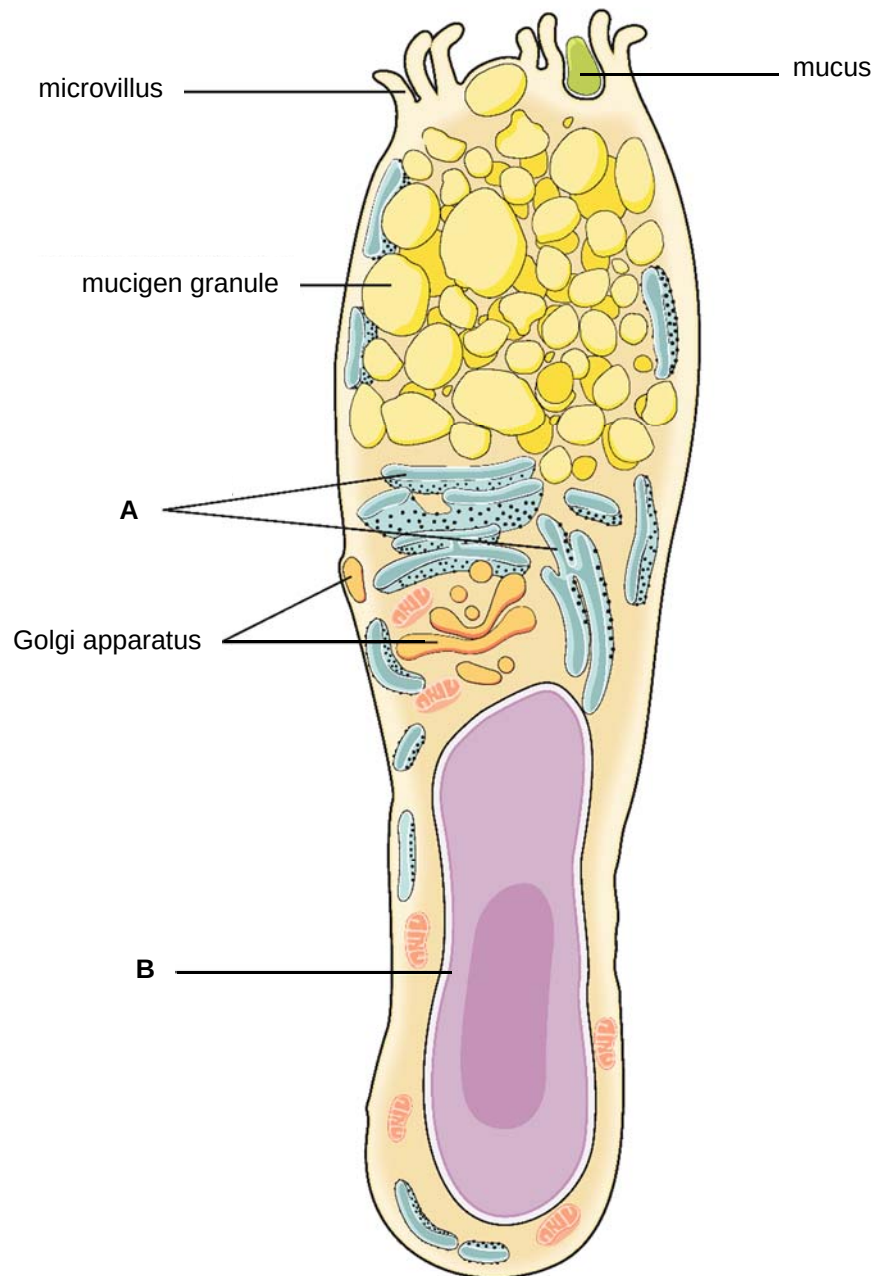


Fig. 1.1

(a) Name the organelles **A** and **B** shown in Fig. 1.1.

A :

B : [2]

- (b)** Explain how organelles **A** and **B** are adapted to allow the goblet cell to secrete mucus out of the cell.

[4]

- (c)** In 1967, Lynn Margulis proposed the theory of endosymbiosis, which states that mitochondria and chloroplasts of eukaryotes evolved from ancient prokaryotes. It proposed that these ancient prokaryotes were engulfed by other bacterial cells and both benefitted from the relationship – this has led to the evolution of eukaryotic cells.

Describe how the similarities between mitochondria and prokaryotic cells support the theory of endosymbiosis.

[Total: 8]

QUESTION 2

Fig. 2.1 is a diagram representing the transport of glucose from the intestinal lumen to the extracellular fluid across an intestinal epithelial cell. Three proteins are labelled **P**, **Q** and **R**.

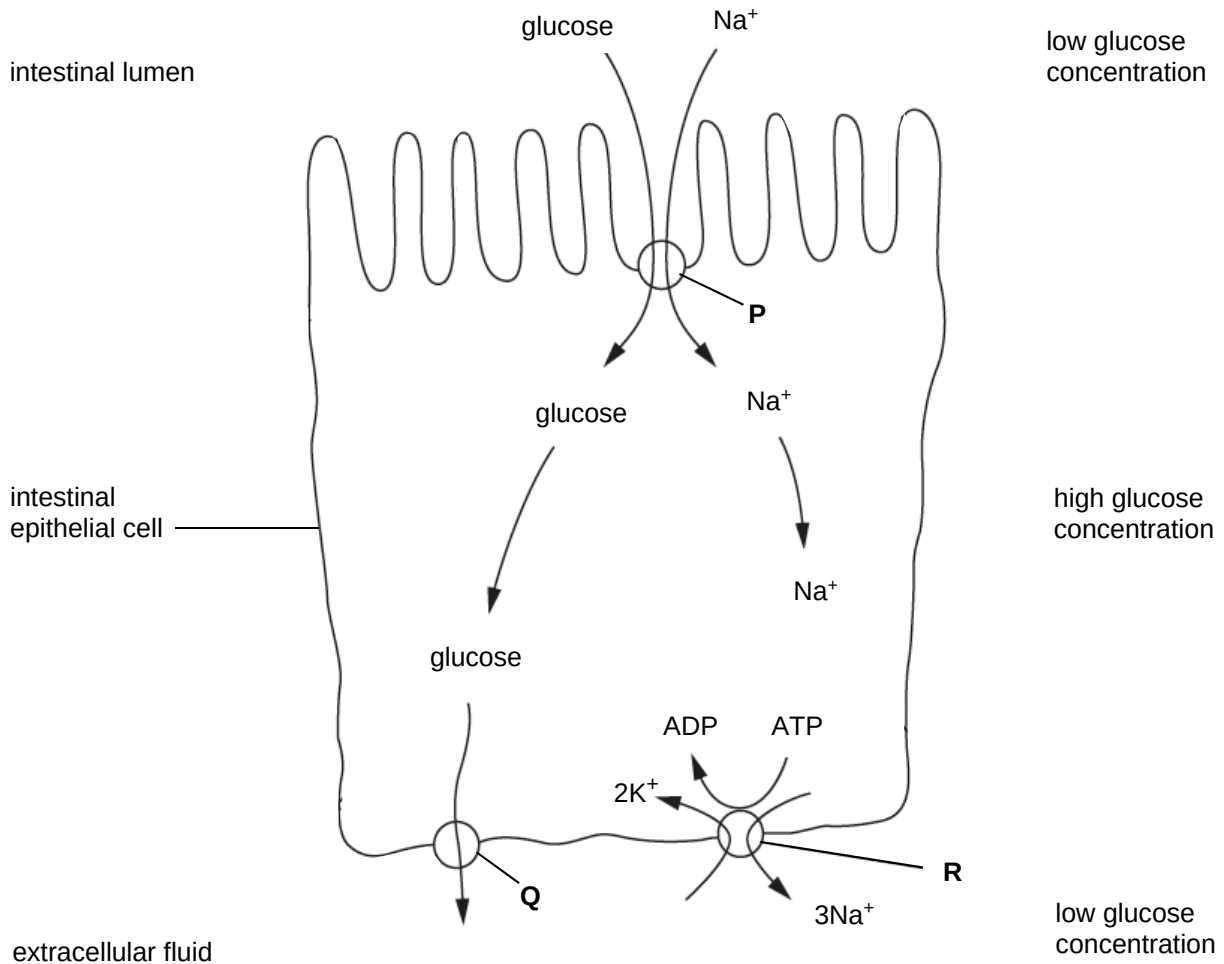


Fig. 2.1

- (a)** Describe what is meant by *facilitated diffusion*.

.....
, [1]

- (b)** Identify which transport protein, **P**, **Q** or **R**, is involved in facilitated diffusion.

....., [1]

- (c) Explain how glucose is transported from the intestinal lumen to the extracellular fluid.

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..... [4]

- (d) Suggest why **R** is necessary in facilitating the transport of glucose across the epithelial cell in Fig. 2.1.

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..... [2]

[Total: 8]

QUESTION 3

Fig. 3.1 shows the gene expression of a ribosomal protein, rS2.

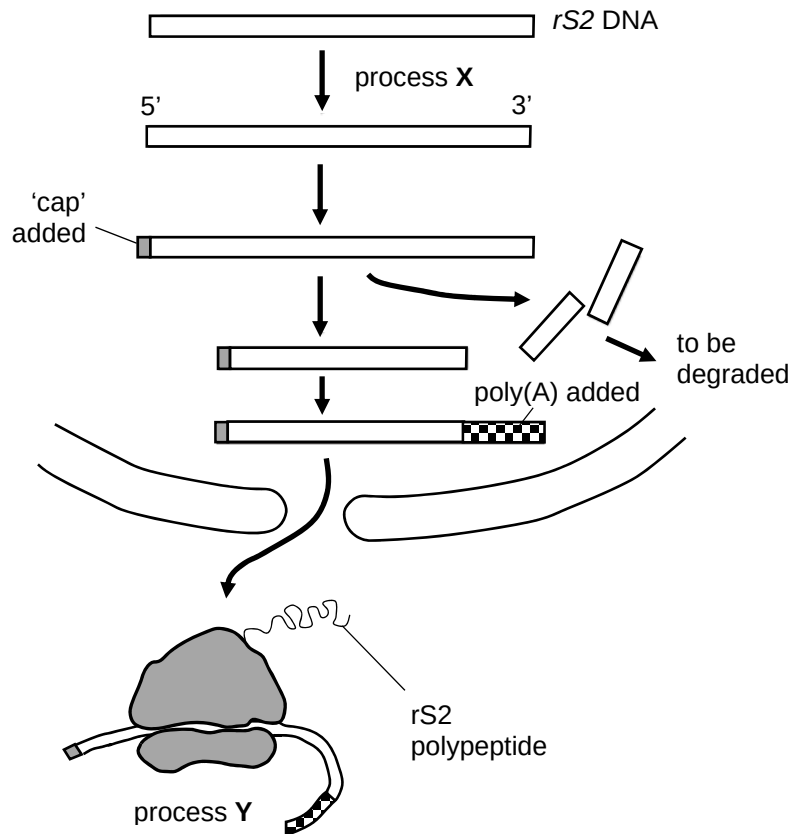


Fig. 3.1

(a) Describe the main structural features of a ribosome.

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..... [2]

(b) State the role of the *rS2* promoter.

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..... [1]

- (c) List three ways by which process **X** differs from process **Y**.

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..... [3]

- (d) Suggest how a mature ribosome is assembled upon the completion of process **Y**.

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..... [3]

[Total: 9]

QUESTION 4

In his book *On the Origin of Species* (1859), Charles Darwin wrote about the high degree of variation among domesticated plants and animals, and their differences from their wild ancestors. One of the animals Darwin studied in depth was the domestic rock pigeon. Pigeon breeding was an ancient pastime, which Darwin himself partook in.

- In birds, the sex chromosomes are referred to as Z and W, rather than X and Y as in mammals.
- The W chromosome has no genes that affect plumage colour.
- The heterogametic sex is the female, not the male.
- Thus the male has two Z chromosomes (ZZ) and the female has one Z and one W chromosome (ZW).

Plumage colour in pigeons is sex-linked, and controlled by three alleles. Ash-red allele (**A**) is dominant to blue (**B**), which is dominant to brown (**N**).

(a) State and explain the meaning of the term sex linkage in this context.

[2]

When an ash-red male was bred with an ash-red female, the majority of the offspring were ash-red, but some of the females were blue. Table 4.1 shows the results of the cross.

Table 4.1

phenotype	number of offspring
ash-red male	7
ash-red female	3
blue female	4

(b)(i) Identify the parental genotypes in this cross.

[2]

ash-red female parent

ash-red male parent

(ii) Using suitable symbols, draw a genetic diagram to explain the results shown in Table 4.1.

[3]

(iii) State the expected phenotypic ratio for the offspring of this cross.

[1]

[Total: 8]

QUESTION 5

Bats are perhaps the most unusual and specialized of all mammals. Together with birds, they are the only extant vertebrates that are capable of powered flight. Bats live in hollow trees, caves and have limited flying range. They have mastered the night skies largely by using echolocation to perceive their surroundings and find their prey in the dark.

- (a) Suggest how the ability to use echolocation may have evolved from an ancestor that did not have that ability.

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[4]

The pipistrelle is the most common species of bat in Europe, the Mediterranean and Asia. It was originally thought that all pipistrelles belong to the same species, *Pipistrellus pipistrellus*. However, in the 1990s, it was decided that there were two species: *Pipistrellus pipistrellus* and *Pipistrellus pygmaeus*.

Table 5.1 shows data for both species.

Table 5.1

species	mean body mass (g)	mean wingspan (m)	range of echolocation call (kHz)	colour
<i>P. pipistrellus</i>	5.5	0.22	42–47	medium to dark brown
<i>P. pygmaeus</i>	5.5	0.21	52–60	medium to dark brown

(b)(i) Name the genus to which the *Pipistrellus pygmaeus* belongs.

..... [1]

(ii) Using the data in Table 5.1, suggest why pipistrelles were originally classified as one species.

.....

 [2]

[Total: 7]

QUESTION 6

Flooding is a severe threat for survival of terrestrial plants, mainly because it severely impedes the gas diffusion rate. Rice (*Oryza sativa*) is the only cereal that can be cultivated in the frequently flooded rivers.

An investigation was carried out into the effect of flooding on the growth of the submerged stems of rice plants. Young rice plants were grown in a container in which the level of water was increased in 10 cm steps, over a period of seven days. The mean length of the submerged internodes (lengths of stem between two leaves) and the concentration of ethylene in the rice stems was measured each day. As a control, rice plants were grown in identical conditions but the water level was kept constant throughout the seven days. The results are shown in Fig. 5.1.

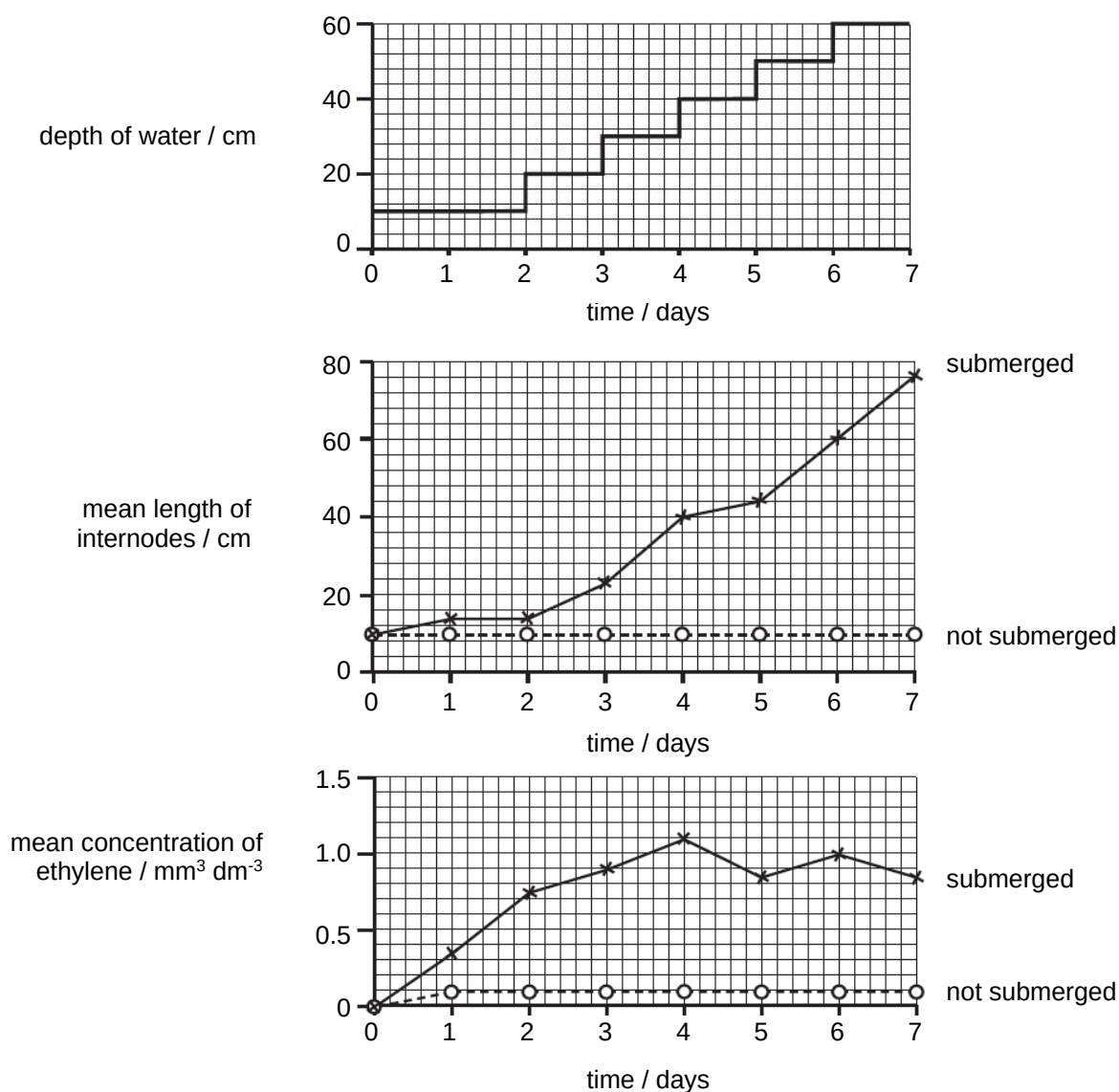


Fig. 6.1

- (a) Describe the effects of increasing water level on the mean length of the submerged internodes and mean concentration of ethylene in rice stems.

[4]

- (b)** Suggest an advantage to the rice plants of the effect that you have described in **(a)**.

[1]

[Total: 5]

--- END OF SECTION A ---

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)** and **(b)** as indicated in the question.

QUESTION 7

(a) Outline the roles of hydrogen bonds in biomolecules. [6]

(b) Discuss the suggestion that all living organisms on Earth depend on phosphate. [9]

[Total: 15]

QUESTION 8

(a) Outline the role of membranes at the surface of and within the cell. [6]

(b) The cell theory is based on a number of principles.

Discuss the extent, in humans, to which these principles can be accounted for by mitosis and the different types of stem cells.

You should consider both individual human and populations of humans from one generation to the next. [9]

[Total: 15]

--- END OF SECTION B ---

--- END OF PAPER---

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